

A negative result on shrinkage estimators in small-N replication

L. Okafor

2026-05-13

Abstract

We attempt to replicate the shrinkage-estimator results of [Brown 2021] under small-N conditions and report a negative finding. Our analysis identifies two likely sources of the original effect: data leakage in the holdout split and a misspecified prior. We release replication code and a corrected reanalysis.

1 Claims

Claim 1 (Claim 1). Brown 2021’s reported shrinkage gains do not survive a leak-free reanalysis.

Replication status: untested.

Claim 2 (Claim 2). The misspecified prior accounts for ~60% of the reported effect size.

Replication status: untested.

2 Notes

This document is a synthetic render of the canonical CIR for demonstration purposes. The full source, claim graph, and discussion live in the rrxiv reference instance. See the rrxiv protocol repository for the schema.